Crypto Lab 1 - Shishir Vuppala

Challenge 2: One-Time-And

flag: `cs409{1f_1t_5h0w5_up_it_w4s_th3r3}`

Solution

There are two cases to consider

1. if the bit to encrypt is a 0 then no matter what the key is $0 \wedge k = 0$ and so in all the generated ciphertexts there will be a 0 in that position.
2. if the bit to encrypt is a 1 then whatever the key value is at that index is what we get since $1 \wedge k = k$. Since we have 20 ciphertexts which all have different keys, the probability that there is no key with a 1 in that spot is astronomically low ($\frac{1}{2^{20}}$) and so almost certainly there will be atleast 1 generated cipher text with a 1 in that spot

Thus, this points to doing bitwise OR of all the generated cipher texts, since if all ciphertexts have a 0 in that position then flag will be 0 almost certainly and if there is even a single 1 in the ciphertexts at that position then the original flag had a 1.

Code

The script used to solve this challenge is written in `decrypt.py`